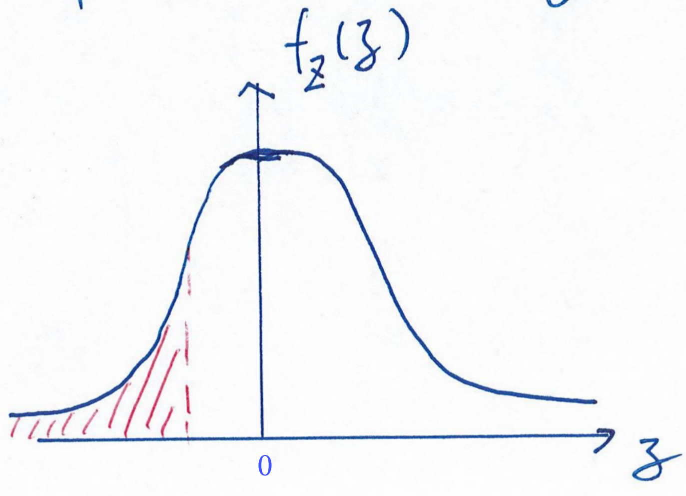
The normal distribution

Def. We say that Z is distributed standard normal if

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in (-\infty, \infty)$$

$$F_Z(z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \, dz$$

has no closed form solution and it is computed numerically.



Def. Let $X = \sigma \cdot Z + \mu$, where $\sigma > 0$ and $\mu \in \mathbb{R}$. Then X follows a normal distribution with PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in (-\infty, \infty).$$

We write $X \sim N(\mu, \sigma^2)$.

Equivelantly, if $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim N(0,1)$ which is the standard normal distribution.

(iii) $M_Z(t) = E(e^{Zt})$

$$= \int_{-\infty}^{\infty} e^{zt}\,\frac{1}{\sqrt{2\pi}}\,e^{-\frac{z^2}{2}}\,dz$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\,e^{-\frac{z^2-2zt+t^2-t^2}{2}}\,dz$$

$$= \overset{\to 1}{\cancel{\int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}}\,e^{-\frac{(z-t)^2}{2}}}_{\text{PDF of } N(\mu,\,\sigma^2)}\,dz}} \cdot e^{\frac{t^2}{2}}$$

$$\mu = t, \quad \sigma^2 = 1$$

$$= e^{t^2/2}, \quad t \in R.$$

- $E(Z) = \frac{d}{dt} M_Z(t) \Big|_{t=0}$

$$= e^{t^2/2} \cdot t \Big|_{t=0}$$

$$= 0$$

- $E(Z^2) = \frac{d^2}{dt^2} M_Z(t) \Big|_{t=0}$

$$= e^{t^2/2} \cdot t^2 + e^{t^2/2} \Big|_{t=0}$$

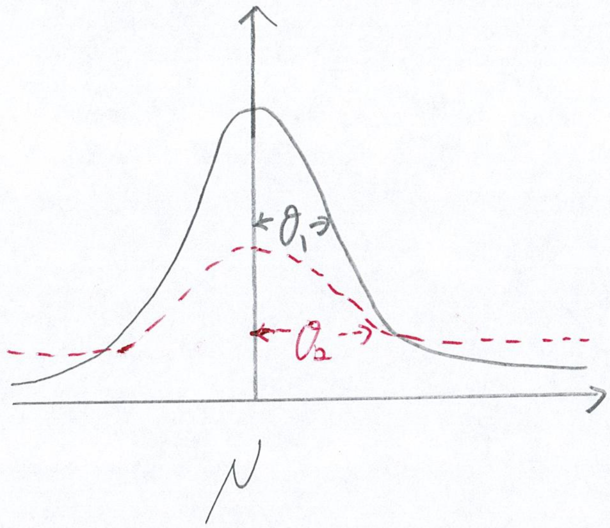
$$= 1$$

- $Var(Z) = 1 - 0^2 = 1$

- $E(X) = E(\sigma \cdot Z + \mu) = \mu$

- $Var(X) = Var(\sigma Z + \mu) = \sigma^2 \cdot Var(Z) = \sigma^2$

(i) If $\sigma_1 < \sigma_2$,



(ii) $M_X(t) = E(e^{Xt})$

$= E(e^{(\sigma \cdot Z + \mu)t})$

$= E(e^{\sigma \cdot t \cdot Z} \cdot e^{\mu t})$

$= e^{\mu t} \cdot M_Z(\sigma \cdot t)$

$= e^{\mu t} \cdot e^{\frac{\sigma^2 \cdot t^2}{2}}$

$= e^{\mu t + \frac{\sigma^2 t^2}{2}}$, $t \in R$.

III THM. Let $X_i \sim N(\mu_i, \sigma_i^2)$ which are independent. Also let

$$S = X_1 + \cdots + X_n \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

$$M_S(t) = \prod_{i=1}^n M_{X_i}(t)$$

$$= \prod_{i=1}^n e^{\mu_i \cdot t + \frac{\sigma_i^2 \cdot t^2}{2}}$$

$$= e^{\left(\sum_{i=1}^n \mu_i\right)t + \left(\sum_{i=1}^n \sigma_i^2\right)\frac{t^2}{2}}$$

$$= e^{\mu^* \cdot t + \frac{\sigma^{*2} t^2}{2}}, \quad t \in R$$

MGF of $N(\mu^*, \sigma^{*2})$

where $\mu^* = \sum_{i=1}^n \mu_i$

$$\sigma^{*2} = \sum_{i=1}^n \sigma_i^2$$

III Ex. Consider $X_i \sim N(19{,}400,\ 5{,}000^2)$

which are independent.

Let $\bar{X}_{25} = \frac{1}{25} \sum_{i=1}^{25} X_i$.

Find $P(\bar{X}_{25} > 20{,}000)$. ← solve this by Φ table.

$$P\left(\frac{1}{25} \sum_{i=1}^{25} X_i > 20{,}000 \right)$$

$$= P\left(\sum_{i=1}^{25} X_i > 25 \cdot 20{,}000 \right)$$

$$= P\left(\frac{S - E(S)}{SD(S)} > \frac{25 \cdot 20{,}000 - 25 \cdot 19{,}4000}{\sqrt{25 \cdot 5{,}000^2}} \right)$$

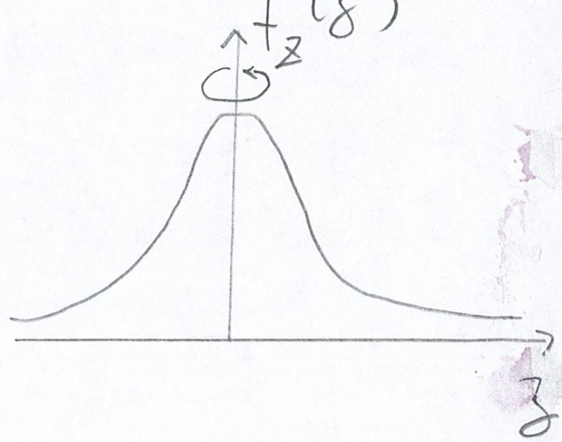
where $S = X_1 + \cdots + X_{25} \sim N\left(\sum_{i=1}^{25} 19{,}400,\ \sum_{i=1}^{25} 5{,}000^2 \right)$

$= P(Z > 0.6)$

$= 1 - P(Z \leq 0.6)$

$= 1 - \Phi(0.6) = 0.2743$

- $-Z \overset{D}{=} Z$



- Let $X \sim N(\mu, \sigma^2)$

$-X = -(\sigma \cdot Z + \mu) = \sigma \cdot (-Z) - \mu$

$\overset{D}{=} \sigma \cdot Z - \mu \sim N(-\mu, \sigma^2)$

- $X_i \sim N(\mu_i, \sigma_i^2)$, then

$X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$